Shuffle-Based Set Membership Proofs

I. Motivation

Set membership proof is a fundamental primitive in cryptographic systems, with applications in authentication, privacy-preserving payments, and access control. On the other hand, verifiable shuffle protocols (e.g., Bayer–Groth) provide efficient zero-knowledge proofs that a list of ciphertexts is a permutation and re-randomization of another. These primitives are widely used in e-voting and anonymous communication systems.

This project explores whether shuffle-based techniques can be systematically leveraged to construct set membership proofs, with a focus on achieving strong privacy guarantees and composability with shuffle-based systems.

II. Problem Statement

The set membership proof is defined as follows. Given a (potentially secret) set $S = \{y_1, \ldots, y_n\}$ and a secret value x, the prover tries to convince the verifier that $x \in S$ without further information regarding either x or S. Here we assume the prover learns both x and S.

A high-level idea on how verifiable shuffling may be used in set-membership proof is as follows:

- Encode elements of S using a commitment to obtain a list of commitments. Similarly, encode x using a commitment scheme
- Perform verifiable shuffling such that the commitment of the element that is equal to x is now on the top of the list
- Perform equality proof that the commitment on the top of the shuffled list and the commitment of x are commitments of the same value.

A. Alternative Formulation via Vector Commitments

While the above approach relies on explicit shuffling of a list of commitments, we consider an alternative formulation that avoids materializing the shuffle explicitly. Instead of committing to each element of S individually, we commit to an ordered vector $\mathbf{v} = (v_1, \ldots, v_n)$ using a Pedersen-style vector commitment:

$$C = h^r \cdot \prod_{i=1}^{n} g_i^{v_i}.$$

The prover aims to convince the verifier that:

- The vector $\mathbf{v}$ is a permutation of the set S,
- The first coordinate satisfies $v_1 = x$.

This formulation can be interpreted as committing to a hidden ordering of S in which the secret value x occupies a fixed position. In contrast to the shuffle-based approach, the permutation is implicit in the committed vector.

III. Technical Challenges

Several challenges arise in making this approach sound and efficient:

- **Hidden Permutation Proof.** A key challenge in the vector commitment approach is to prove in zero knowledge that the committed vector corresponds to a permutation of the public set S. Unlike standard shuffle arguments, where permutations are proven over explicit lists, the permutation here is implicit in the committed vector. Designing efficient zero-knowledge proofs for this property is non-trivial.
- **Trade-off Between Explicit and Implicit Shuffles.** The vector commitment formulation improves compactness but removes the direct applicability of standard verifiable shuffle techniques. Understanding the trade-offs between these two approaches is an important research direction.
- **Hybrid Commitment Structures.** Beyond the two extreme approaches of committing each element individually or committing the entire set as a single vector, we propose to investigate hybrid constructions. In such a construction, the set S is partitioned into multiple subsets, each of which is committed using a vector commitment.
  This yields a representation consisting of several vector commitments, each encoding a subset of elements. By adjusting the partition size, this approach provides a spectrum of trade-offs between:
  - the efficiency of shuffle-based proofs (which favor smaller blocks or element-wise commitments), and
  - the compactness of vector commitments (which favor larger blocks).

  A key research question is how to design efficient protocols that combine local shuffle arguments within blocks and global consistency guarantees across blocks while preserving zero-knowledge.

IV. Proposed Methodologies

We recall that a key goal of this project is to evaluate whether shuffle-based approaches provide competitive advantages over existing set membership constructions (e.g., Merkle trees or accumulators), or to identify the regimes in which they are beneficial. The project will proceed in stages:

1) Literature study on existing set-membership proofs
2) Exploration of basic shuffling-based approach
3) Exploration of vector-commitment based implicit shuffling approach
4) Protocol Implementation and basic comparison for feasibility of shuffling-based approaches, including the discussion of applications or use cases where shuffling-based approaches may gain advantages
5) Security and Complexity analysis

V. Deliverables

In this project, we have the following deliverables:

- Report on the proposed solution(s), privacy guarantee, and complexity analysis
- Implementation of the proposed solution(s) and its experimental result compared to other approaches for set membership proof to show the feasibility of shuffling-based approaches.

Rekomendasi untuk implementasi: Rust